

A top-down design flow for MOEMS

Gunar Lorenz^a, Arthur Morris^b, Issam Lakkis^c

Coventor, Inc.

^a3 Avenue de Quebec, Courtaboeuf, 02142, France

^b4001 Weston Parkway, Cary NC, 27606, US

^c215 First Street, Cambridge MA, 02142, US

ABSTRACT

As MOEMS (Micro-Opto-Electro-Mechanical Systems) transition from laboratory curiosities to production products inserted into telecommunication systems, a top down design flow is necessary to minimize time-to-market. We present a structured-custom approach for MOEMS design based on the concept of using parameterized behavioral models as a means to improve and speed up the design process. This approach enables a user to quickly explore a larger design space on the interaction of the behavioral models with their surrounding system. Once the simulation is completed to the designer's satisfaction, a device layout can be output for FEM verification on critical areas or to generate the masks for fabrication.

Keywords: MEMS, MOEMS, optomechanics, free-space optics, gaussian beam propagation, system-level simulation, parametric models

1. INTRODUCTION

MOEMS are emerging from the laboratory to provide unparalleled functionality in optical telecommunications applications. While the ultimate speed of these devices is unlikely to compete with solid-state or electro-optic devices, the transparency that can be achieved with MOEMS contributes negligible degradation to optical channels and thus enables long range flexible all-optical networks. Beam steering has been shown to be the optimum approach for scaling optical cross-connect switching to large port counts and MOEMS have proven to be the method of choice for compact implementations of this concept. Other telecommunication applications include smaller switches, variable attenuators, equalizers, modulators, and polarization and dispersion compensators. Applications outside optical communications for MOEMS include scanning, projection, display, printing, sensing, and data storage¹.

In these optical subsystems, the choice of the "right" MOEMS device for a given application is driven by the MOEMS device interaction with the rest of the system. This includes free-space and guided-wave optical elements, the MOEMS packaging, opto-electronics, and the control system for the MOEMS. To evaluate these interactions, the designer must be able to simultaneously model all of these areas in sufficient detail to enable tradeoffs to be made without requiring such extensive computation that interactive design is impossible. This work describes the models of top down design approach for MOEMS. A comparison between system model calculation and coupled FEM/BEM simulation verifies the feasibility of electromechanical simulation achieving the accuracy of field solving codes at network-type simulation speed.

2. THE STRUCTURED-CUSTOM DESIGN APPROACH

In the structured custom design approach, the system specifications are applied to the integrated high-level system to obtain the required MOEMS subsystem performance. This initial set of MOEMS specifications is used to select a design and fabrication approach. Choices made at this point could include digital vs. analog control, surface vs. bulk micromachining, actuation method (thermal, electrostatic, magnetic, fluidic, etc.), and range of motion required. An initial system model of the MOEMS component is then built from parametric primitives in a schematic. The system model enables the user to quickly explore a large design space based on the interaction of the behavioral models with their surrounding system. For example, cross sensitivity of a device that has multiple degrees of freedom can be evaluated. Or, one can evaluate how resonant frequencies change if the fabrication tolerances change. All of the criteria that define the technological process can be investigated very rapidly but would take days or weeks with a FEM approach.

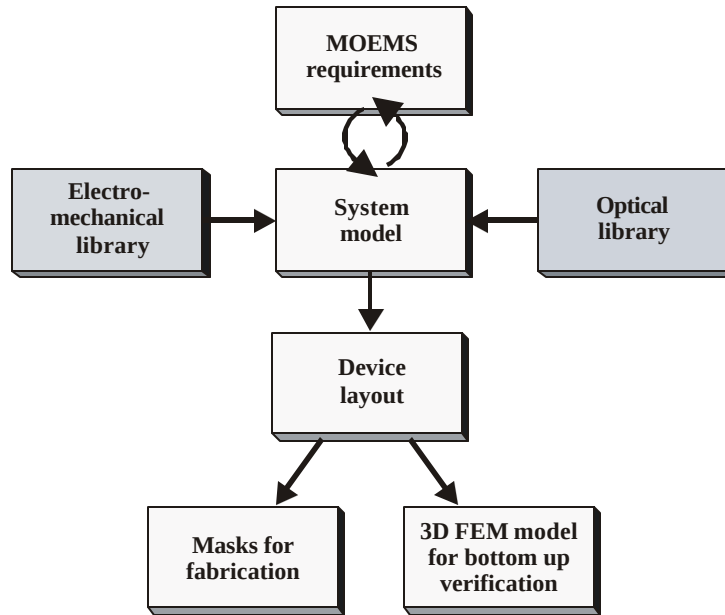


Figure 1: The structured-custom design flow

Once the simulation is complete to the designer's satisfaction, a device layout can then be generated from this high level description in the context of the chosen process. If the design is simple enough, this may be sufficient to tape out for fabrication. The more likely path involves detailed 3-D PDE analysis to verify the desired behavior and make any corrections necessary for interactions not included in the parametric models¹, compare with Figure 1.

3. ELECTROMECHANICAL COMPONENTS

The electromechanical components for the MOEMS design flow are part of Coventor's electromechanical library (see Figure 2). The electro-mechanical library has been developed to allow rapid creation of 6-DOF electromechanical models of micro machined devices such as mirrors, resonators, accelerometers and gyros^{2,3}.

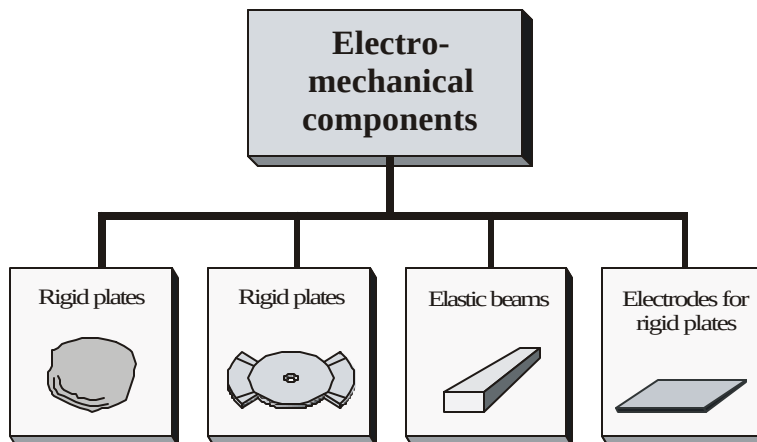


Figure 2: Electromechanical components for the MOEMS design

The elements of the electromechanical library provide quite general geometry, are fully parametric in material properties and dimensional scaling and provide simulation accuracy comparable to FEM (but orders of magnitude faster) in a system or circuit simulator⁴. A particular emphasis has been placed on the modeling of geometrical properties caused by the manufacturing process such as stress related surface bending or sidewall angles. The influence of manufacturing tolerances, surface stress and the interaction of different system components can be investigated with just one model using the frequency or transient analysis tools of commercial network simulators such as SABER from Avant!.

1. The mechanical interface

Each library component represents the electromechanical behavior of a certain substructure in the device. The substructure's geometry and position in the common substrate coordinate system is definable with the parameters of the corresponding schematic symbol. On the schematic level, the different symbols are mechanically linked by six wires. This "mechanical wires" are connected to the symbols standard interface consisting of the six mechanical ports x , y , z , rx , ry , and rz see Figure 3.

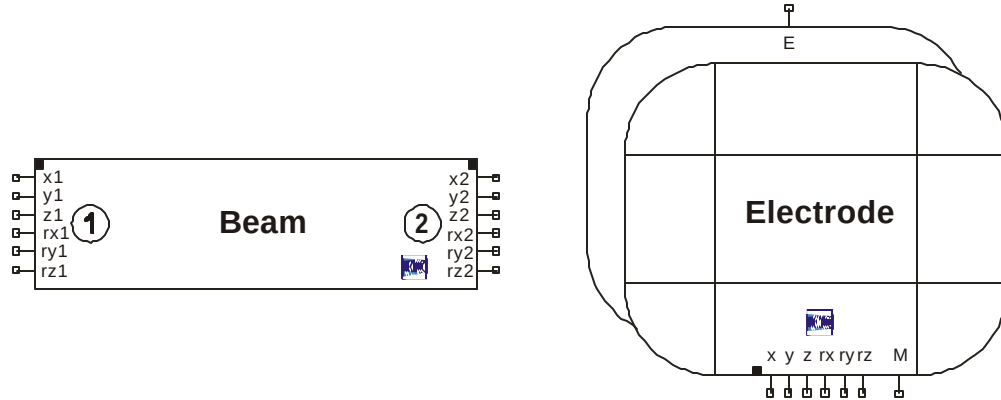


Figure 3: Example symbols from the electromechanical library

The connected mechanical wires refer to a movable point in space called the mechanical "knot". The cross variables (voltages) of the wires describe the knots displacement and rotation respectively. The through variables (currents) are interpreted as the forces and torques acting on the knot. In a mechanical network model, the structure's degrees of freedom are reduced to motions and rotations of one or more knots. The number of knots (mechanical wires) needed to describe a mechanical structure correctly depend on its complexity and the chosen level of abstraction.

The behavior description for each library component follows the same basic principle and can be divided into two steps. In the first step, the substructure's current position is calculated by taking the displacement and rotation of the corresponding knot/knots as an input (beams are linked to two different knots, see Figure 3). All needed coordinate transformations are provided by the model. In the second step, the substructure's force and torque contribution at the knot(s) is calculated based on the inertial, elastic or electrostatic boundary conditions.

2. Rigid plates

All parts of the device geometry which add no significant elasticity to the structure are considered as rigid plates. The geometry of rigid structures can be specified using up to 40 different standard segments. Each of the segments is determined by a set of parameters defining its material, geometry and position on the substrate. Four different segment types are available, consisting of rectangular and pie shaped plates as well as curved and straight comb fingers. In addition to the geometrical dimensions, each plate segment also includes properties to specify its perforation and a fabrication caused curvature. Curvature is internally approximated by vertical offsets of elementary cells, the internal building blocks of a rigid plate (compare with Figure 4).

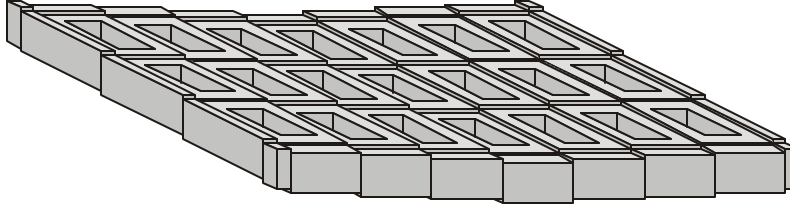


Figure 4: Rectangular plate segment

The mechanical properties of a rigid plate are defined by its mass, its center of mass, and its moments of inertia. The rigid plate model calculates all three mechanical values automatically, before the actual simulation starts. During the simulation, forces and torques at the plates are calculated according to Newton's law, and the Euler equations, taking into account tiny motions on the plates relative to the substrate as well as large substrate motions. Large substrate motions (accelerations and angular rates) can be applied to a separate global symbol called the "R" Figure 8b.

3. Beams

Flexible structures like plate suspensions are represented using three-dimensional beam elements. With the beam component from the parametric library, the user has access to different underlying models. By choosing the appropriate model, the user can decide whether the beam's inertia and gas damping are neglected or not. Furthermore, each beam model is equipped with a set of parameter switches to adjust for the beam models complexity. Switches for large beam motion, substrate motion and nonlinear bending are available.

Each beam model consists of 1 to 4 beam segments. The behavioral description of a beam segment is based on the same linear beam theory that is commonly used in beam elements of mechanical FE-codes. The theory of linear beams elements is founded on the following assumptions:

1. The beam end faces are rigid (and firmly linked with the connected knot.)
2. The beam end faces are perpendicular to the neutral (central) axis.
3. Torsion, bending and expansion are considered to be independent.
4. The neutral axis is described by a cubic polygon in space, yielding a second order approximation of beam bending.
5. Torsion and expansion are assumed to be linear along the neutral axis.

Taking these assumptions into consideration, the 12 mechanical degrees of freedom of a beam element (6 for each beam face) can be described by a 12x12 mass and stiffness matrix $\mathbf{M}$ and $\mathbf{K}$ ⁵:

$$\mathbf{F} = \mathbf{M}\ddot{\mathbf{x}} + \mathbf{D}\dot{\mathbf{x}} + \mathbf{K}\mathbf{x}$$

Forces and torques at the beam ends are represented by the vector $\mathbf{F}$. Linear and angular displacements of both beam ends are comprised in the 12 components of the vector $\mathbf{x}$. Unlike FE-codes, the acceleration vector $\ddot{\mathbf{x}}$ of the library beam takes large motions of the substrate into account, giving it an implementation of second order effects like the Coriolis force. By separating tiny on chip displacements from large substrate motions, the presented approach enables the simulation of 3D on-chip displacements allowing arbitrary device motions in space.

Despite the internal use of the linear beam theory, the parametric beam can also include nonlinear bending effects by setting the parameter switch *nonlinear_bending* to on. In that case the coupling between bending and expansion is incorporated by using the endpoints of the beam segment as the references for its current length, as seen in Figure 5. By superimposing cross coupling between bending and expansion the implemented beam models react to large bending with an appropriate lengthening of its neutral axis. The accuracy of this approximation depends on the beam's distortion and the number of beam segments used⁶

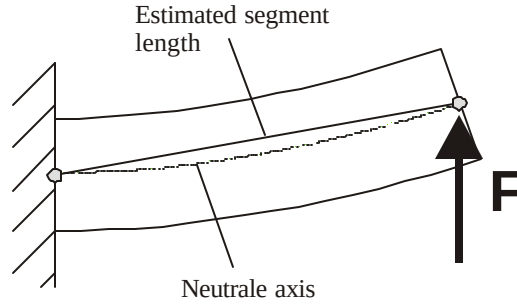


Figure 5: Estimation of the beam segment length

The number of segments used for each beam model can be defined by a model parameter. By choosing the number of segments, the user has control over the approximation used for the neutral axis of the beam. Considering that each beam element approximates bending with a second order polynomial, the number of beam segments can also be seen as the number of correct natural frequencies that are taken into account during the simulation.

4. Electrodes

The electrode model calculates the electrostatic forces and the capacitance at the plate above an electrode according to the applied voltages and the position/orientation of the plate. The capacitance value is internally used to model the electrical behavior of the model's two electrical pins E and M, compare with Figure 3. The user can access the internal capacitance value via an internal node.

The electrostatic force and capacitance calculation is based on the simplified view of the electrostatic field between rigid plate and electrode. For its calculation two different planes are introduced; the first located at the lower side of the rigid plate and the second on the upper side of the electrode see Figure 6.

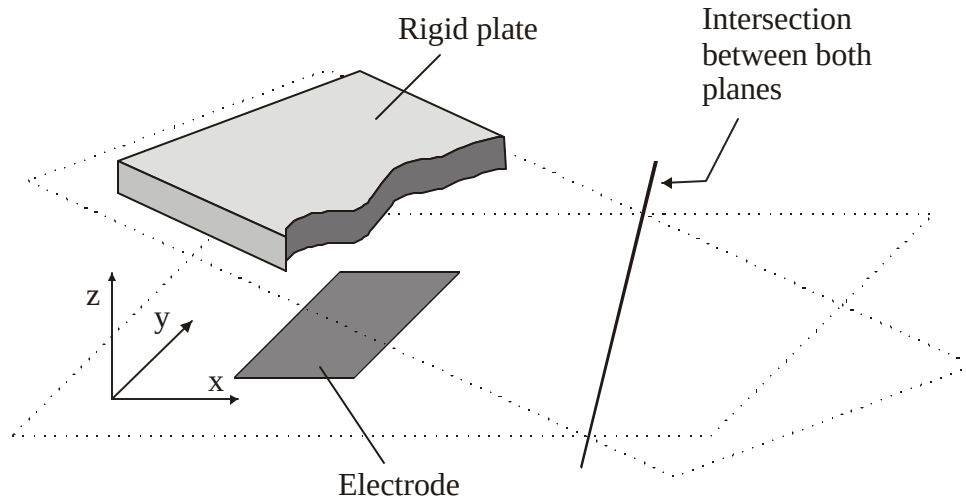


Figure 6 Plane definition for the analytical electrode model

Like the rigid plate itself, the upper plane is following the motion and rotation of the corresponding mechanical knot while the electrode plane stays parallel to substrate. The electrostatic field lines between the rigid plate and the electrode are considered to be arcs with the radius d and the intersection angle φ , compare with Figure 7.

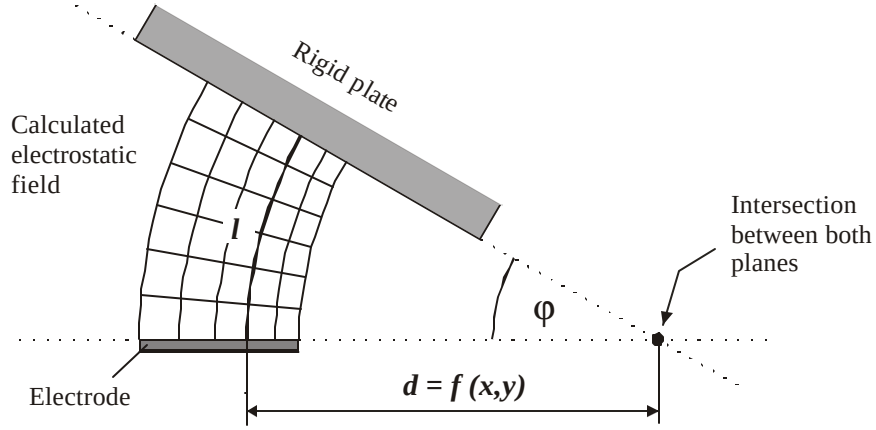


Figure 7 Considered electrostatic field geometry

The length of each field line l is given by d times ϕ . Hence, the capacitance C and the electrostatic force F_n acting on the rigid plate can be calculated by integrating over the electrode's area A :

$$C = \iint_A \frac{e}{dj} dA$$

$$F_n = \iint_A \frac{U^2 e}{2(dj)^2} dA$$

Where U stands for the voltage between the electrode and the rigid plate (across variables of the wires connected to the two electrical ports M and E compare with Figure 3). The electrostatic force F_n acts normal on the lower side of the rigid plate and is the cause of the electrode model's force and torque contribution to the corresponding mechanical knot.

5. Layout extraction

Each symbol in the electromechanical library represents two different “views” of the same mechanical substructure, a behavioral view as described in the previous sections and a layout view in the form of layout generators. Each view is driven and characterized by the same symbol parameters in the schematic level.

The Schematic is converted to layout by importing the corresponding SABER netlist into Coventor's layout editor and 3d model builder MEMSDesigner⁷. The layout extractor parses the netlist file for interpretable model components. If such a component is found, all geometry related properties are extracted from the netlist and passed over to the appropriate layout generator. The layout generator uses these parameters to visualize specific information in one or more layers.

All generators of the layout extractor have an additional user generator. These user generators are customizable extensions for the default generators, which can be used to add user specific layout information and/or additional layers, e.g. specific anchor geometries.

The layout view of a model serves two different purposes. The first and maybe most important is to make photo masks so that the device can be manufactured. Second is to build a solid (or 3d model) that can be meshed and analyzed in a FEM driven verification step.

6. Comparison with a coupled FEM/BEM approach

The following example compares the results of a parametric system model with the numerical results from a simulation where the finite element method (FEM) was used for mechanics and the boundary element method (BEM) for electrostatics. The simulated device is a simple mirror suspended with two torsion beams; see Figure 8(a). The corresponding SABER Sketch model, consisting of two beams, two electrodes and a rigid plate can be seen in Figure 8(b). All five components are linked by six mechanical wires referring to the six degrees of freedom of the common mechanical knot (located in the center of the plate).

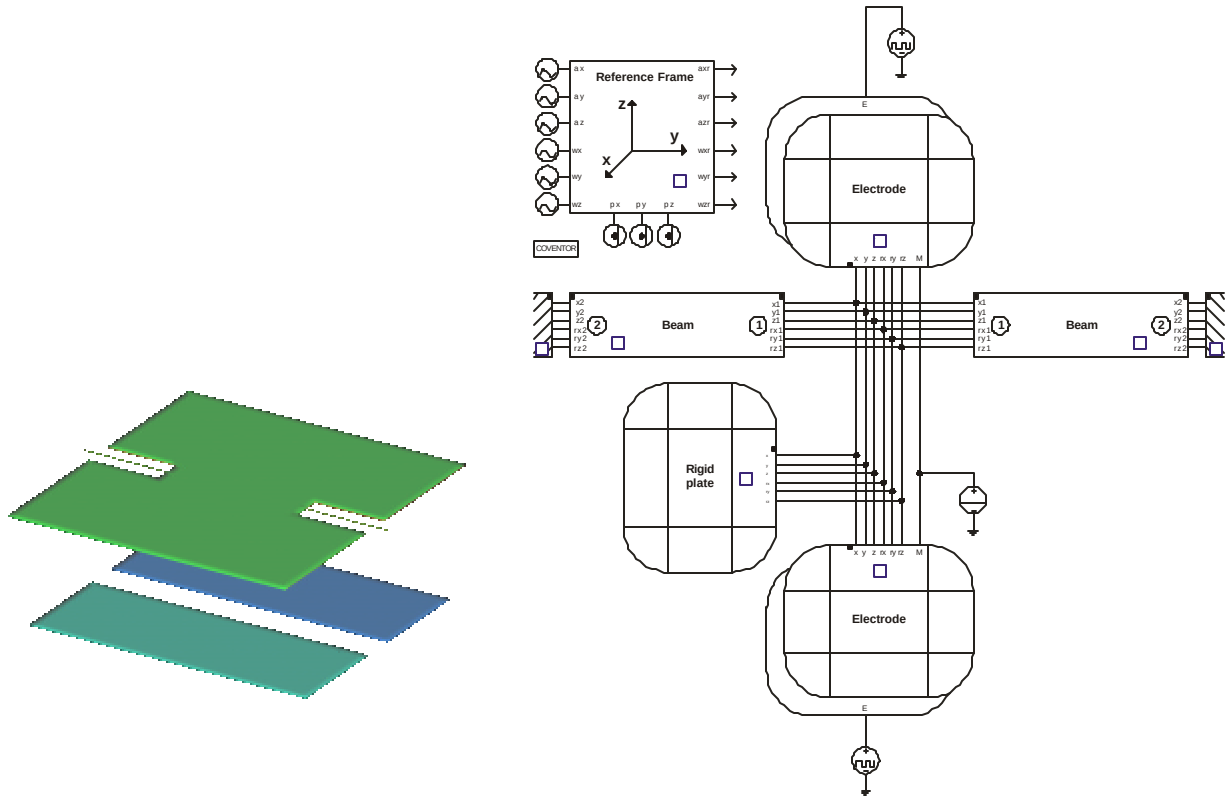


Figure 8: (a) 3d model of the mirror geometry and (b) System model of mirror

For verification the pull-in angle and the vertical displacement of the mirror was simulated with a dc-transfer simulation. The voltage was applied to only one electrode. The second electrode was grounded. The resulting tilting angle (across variable of wire rx) and the vertical motion of the plate center (across variable of wire z) are displayed in Figure 9 and Figure 10.

In a second step the solid model (Figure 8a) was built by extracting the device layout into Coventor's MEMSDesigner. After meshing the solid model with 682 and 1176 26-node bricks, respectively the structure was simulated with Coventor's coupled FEM/BEM solver CoSolve; see also Figure 9 and Figure 10. The accuracy of the coupled FEM/BEM simulations converges towards the results of the network model by increasing the number of elements. However, the most dramatic advantage of the presented simulation technique reveals by comparing the simulation times of the two different approaches, see Table 1.

Table 1: Simulation time of the System model compared with FEM/BEM (750 MHZ PC with 1G RAM)

Simulation approach	Simulation time for the DC transfer simulation (no contact)
FEM/BEM using 682 elements	2.5 hours
FEM/BEM using 1176 elements	8 hours
SABER system model	< 2 seconds!!!

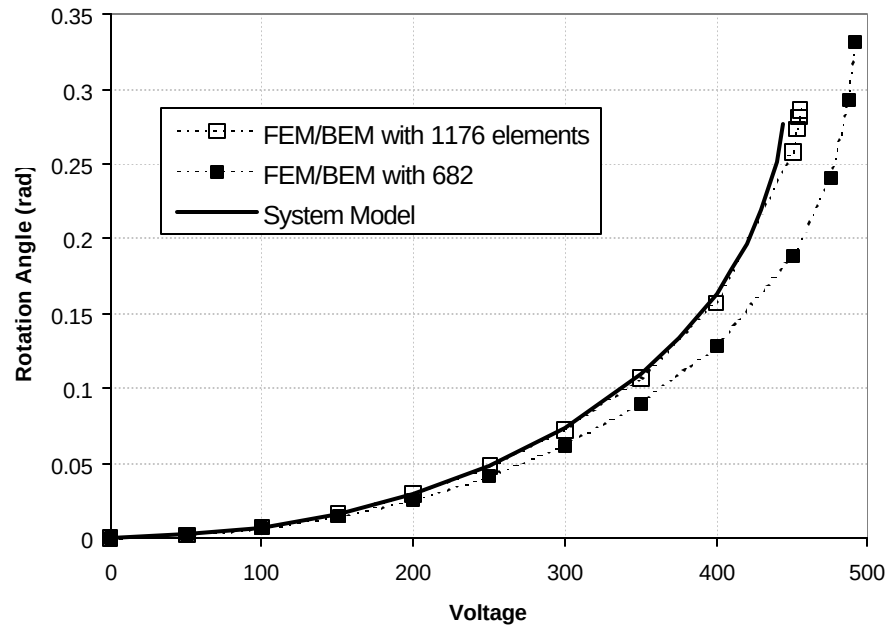


Figure 9: Verification of the pull-in rotation angle (node rx)

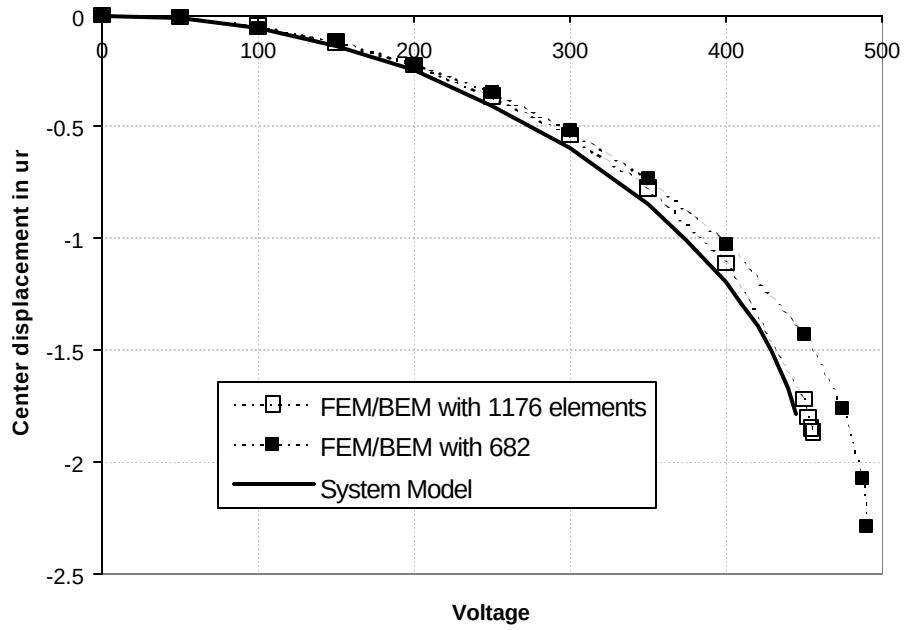


Figure 10: Verification of the pull-in displacement of the mirror center (node z)

4. OPTICAL COMPONENTS

One previous barrier to the integration of optics and MEMS devices was the lack of optical system-level simulation capability to enable the co-design of MOEMS simultaneously with embedding control electronics and macro-optical components and packaging. We have addressed this issue by creating a 6 DOF Gaussian beam optics model library (OptoLib)⁸ to work hand-in-hand with our electromechanical models, see Figure 11.

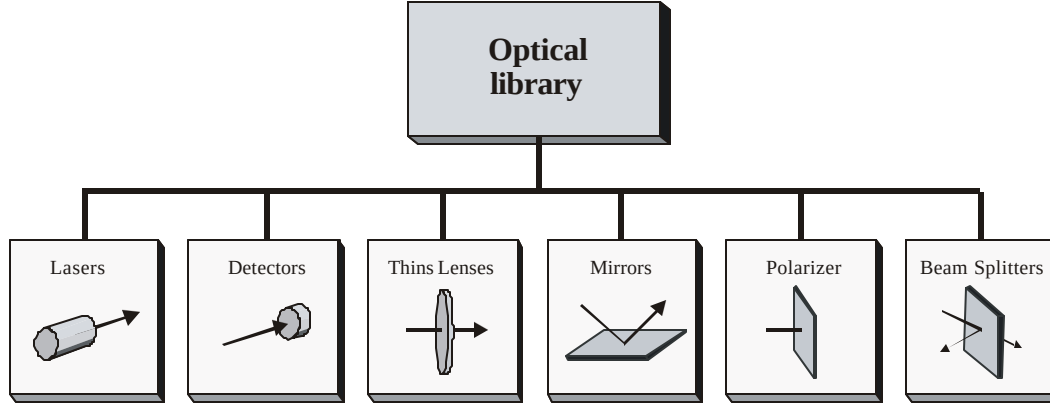


Figure 11: Optical components for MOEMS design

The optical components are connected via a standard optical interface representing a gaussian beam see Figure 12. Eleven wires are necessary to define the beams characteristics: the beam power, the wavelength λ , the Rayleigh range of the beam z_0 , the position of the previous component along beam z-axis z_s , the beam position in the reference frame both translational x, y, z and rotational ρ, θ, γ and finally the phase of beam at beam waist ϕ .

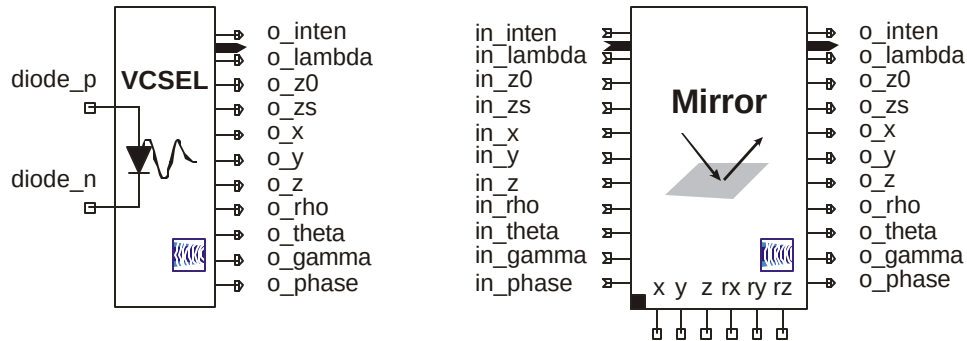


Figure 12: Example symbols from the optical library

The optical system models use a Gaussian beam analysis, which is based on the following assumptions:

- The light is assumed to be in gaussian beams.
- Initial models use circular beams without astigmatism. This is not a requirement but simplifies the initial devices. Future devices such as cylindrical lenses and stripe lasers will include these effects.
- Light propagation is assumed instantaneous with phase change. This is acceptable if none of the elements are moving at a substantial fraction of the speed of light and that optical transients are slow compared to the optical transit time through the system.
- The optical elements do not produce mechanical feedback. Initial elements are connected to mechanical models that provide this information. Optical effects such as radiation pressure or optical heating are not included yet.

The advantage of using Gaussian beam analysis is the fast computational speed in which light is modeled and propagated, allowing for interactive system-level design. For some micro-systems, diffractive effects can dominate, and our Gaussian propagation will not always be accurate⁹. For these systems, other optical methods, such as scalar or vector diffraction propagation, must be used¹.

The position and orientation of each of the optical elements is completely arbitrary. The dynamic position is set by the same 6 DOF connections used in the electromechanical model libraries. This enables the designer to create complex optical systems including the motion of the MOEMS themselves. The library provides for the treatment of astigmatic and elliptical Gaussian beams and preserves both phase and polarization.

5. MOEMS EXAMPLE

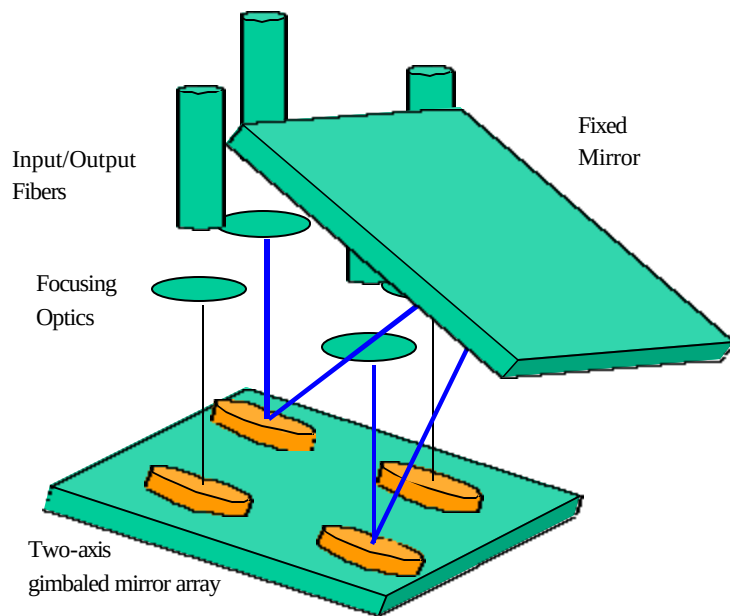


Figure 13: Large Scale Optical Cross Connect Drawing

The overall system containing the domains of gaussian beam free-space optics, 6-DOF electromechanics, and mixed signal ICs must be simulated by the system architect to reasonable accuracy. Both qualitative and quantitative tradeoffs among the system sub-domains shall be demonstrated with the following example. Consider an optical cross-connect utilizing two-axis mirrors for beam steering. One implementation of such a system is shown in Figure 13. This involves detailed design of the MEMS two-axis mirrors, optical system design, optical packaging and alignment, and control electronics for the mirror movement. We begin with a concept for a mirror. As a simple example, consider a square two-axis mirror with straight torsional springs. This was rapidly constructed using our parametric library of electromechanical primitives and is shown in Figure 14(a). For any given set of parameters, the corresponding layout and solid model can be generated; either for visualization or to continue to meshing for FEM or BEM simulations. For a given set of parameters, these are shown in Figure 14(b).

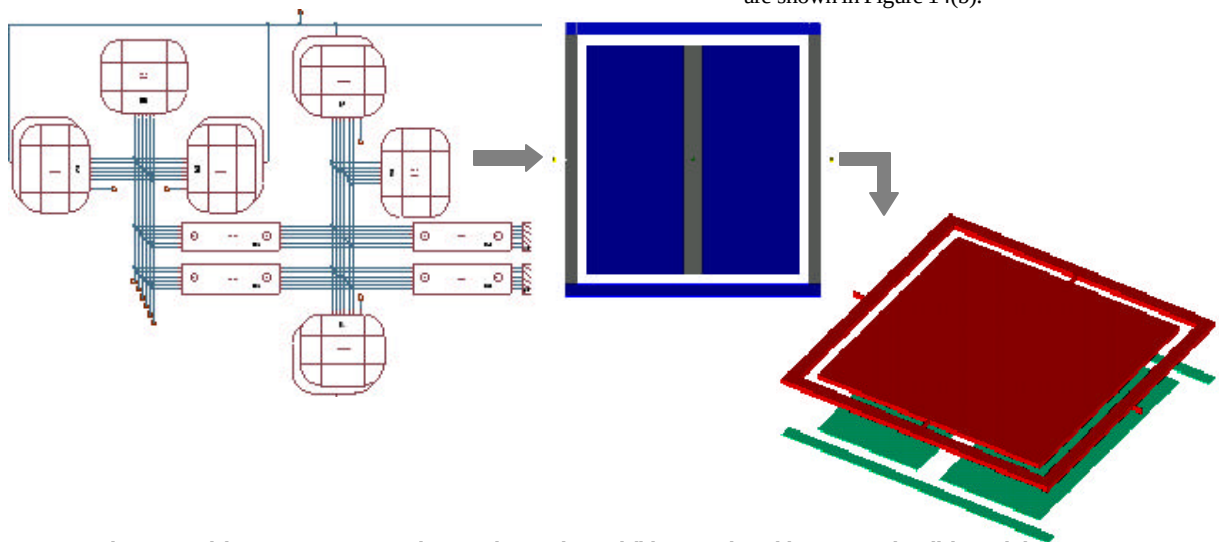


Figure 14: (a) Cross connect mirror schematic and (b) associated layout and solid model

A hierarchical symbol for this mirror model is then created so that it can be placed several times and so that commonly varied dimensions are captured as parameters at the higher level. These mechanical mirrors are then used together with optical elements and electronic elements to create a schematic of the example cross connect system. This schematic is shown in Figure 15(a). The optical beam starts from a laser at the upper left, is focused by a lens, and then reflected by a MEMS mirror. Note the 6-DOF connections between the electromechanical mirror and the optical mirror. Next the large fixed mirror reflects the beam. At this point, we run the same beam information to two different locations in the system. This beam is sampled both by the intended second MEMS mirror for evaluation of channel loss and alignment sensitivity as well as another MEMS mirror for evaluation of crosstalk. Since these mirrors are placed parametrically anywhere in the MEMS array, the performance of any combination of mirrors can be studied with this simple schematic. In Figure 15(b), the signal from the intended channel is plotted versus the focal length of the lenses at a sequence of lens positions. These results can be used to minimize the optical loss of the system while simultaneously minimizing the sensitivity to variations. In Figure 15(c), the same signal is plotted as a function of the x-axis actuation voltage at y-axis voltages from 30 to 40 volts. The required mirror electrode voltages for maximum signal are easy to determine and the performance sensitivity to voltage variation can be evaluated.

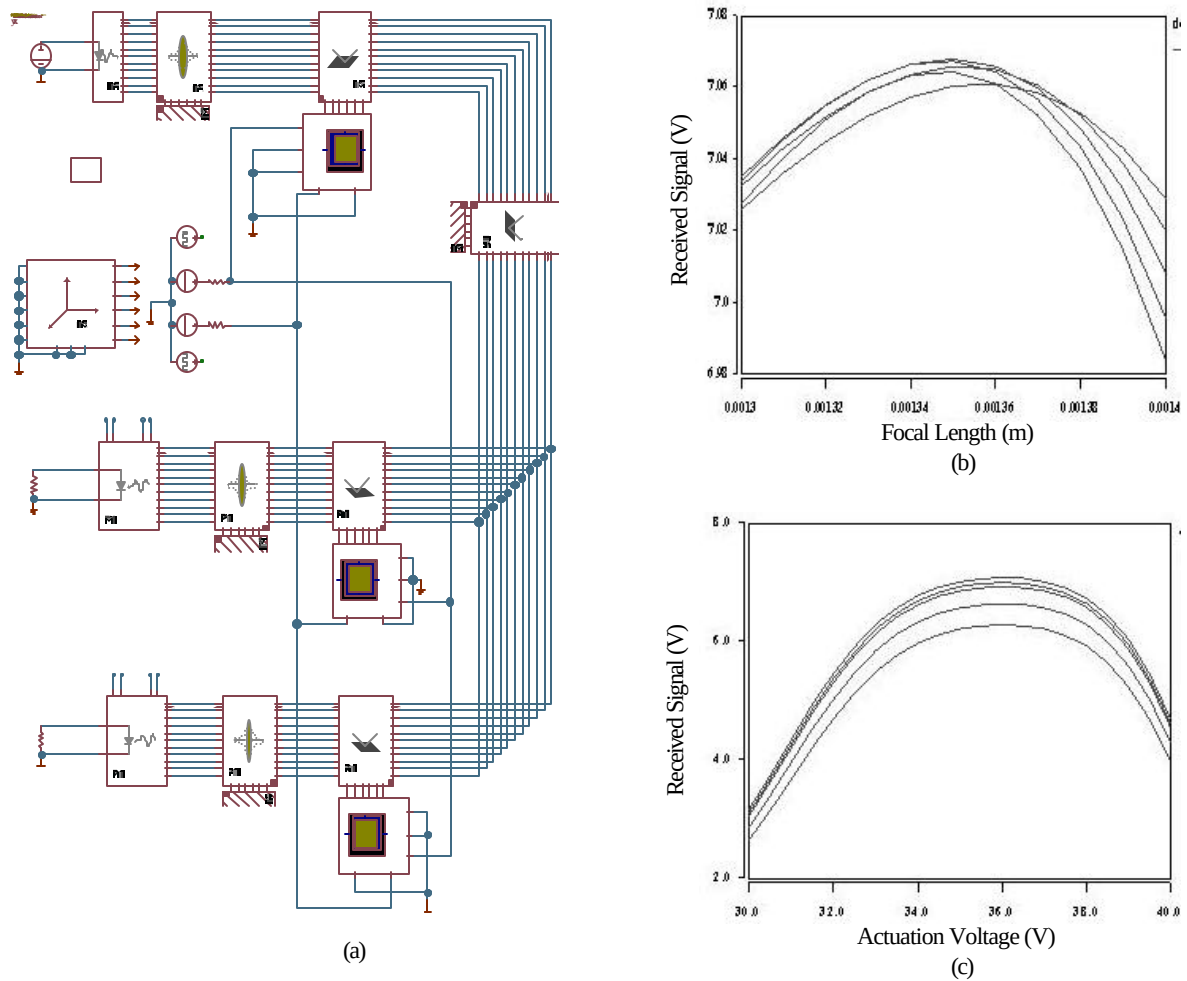


Figure 15: Cross Connect System Schematic (a) and Simulation Results (b and c)

6. CONCLUSIONS

We have presented a comprehensive capability for the design and analysis of optical MEMS systems. Our new structured-custom approach starts with system requirements and uses parametric elements for initial designs that can meet these specifications. Optical, electronic, and mechanical aspects of the systems can be simulated simultaneously, enabling true system level design of applications incorporating optical MEMS. This provides rapid evaluation of the system performance and reserves intense numerical analyses for evaluating second order effects and couplings in a bottom-up verification/design centering step. Comprehensive FEM and BEM analysis capabilities are available for these evaluations and can also be exercised to extract precise behavioral models for final inclusion in system simulations.

REFERENCES

1. Arthur Morris, Stephen Bart, Derek Kane, Gunar Lorenz and Vladimir Rabinovich.: A design flow for MOEMS, SPIE Photonics East 2000
2. Teegarden, D.; Lorenz, G. and Neul, R.: How to model and simulate microgyroscope systems. IEEE Spectrum, July 1998, pp. 66-75.
3. Reinhard Neul, Gunar Lorenz and Stefan Dickmann: Modelling and Simulation of Micro Electromechanical Sensor Systems, Proceedings of the VDE World Microtechnologies Congress MICRO.tec 2000 September 25th - 27th, 2000, Hannover, Germany, Volume 2, pp. 67 - 72.
4. Lorenz, G.: *Netzwerksimulation mikromechanischer Systeme*. Dissertation, Universität Bremen, Shaker Verlag, 1999.
5. Przemieniecki, J. S.: *Theory of Matrix Structural Analysis*. Dover, 1985.
6. Lorenz, G.; Neul, R.: Network-Type Modeling of Micromachined Sensor Systems. *Proc. Int. Conf. on Modeling and Simulation of Microsystems, Semiconductors, Sensors and Actuators*, MSM98, Santa Clara, April 1998, pp. 233-238.
7. Coventor, <http://www.coventor.com>
8. Kurzweg, et al., *Proceedings of Modeling and Simulation of Microsystems*, 768-773, June 2000.
9. Walker and Nagel, "Optics & MEMS," Naval Research Lab, NRL/MR/6336 99-7975.